

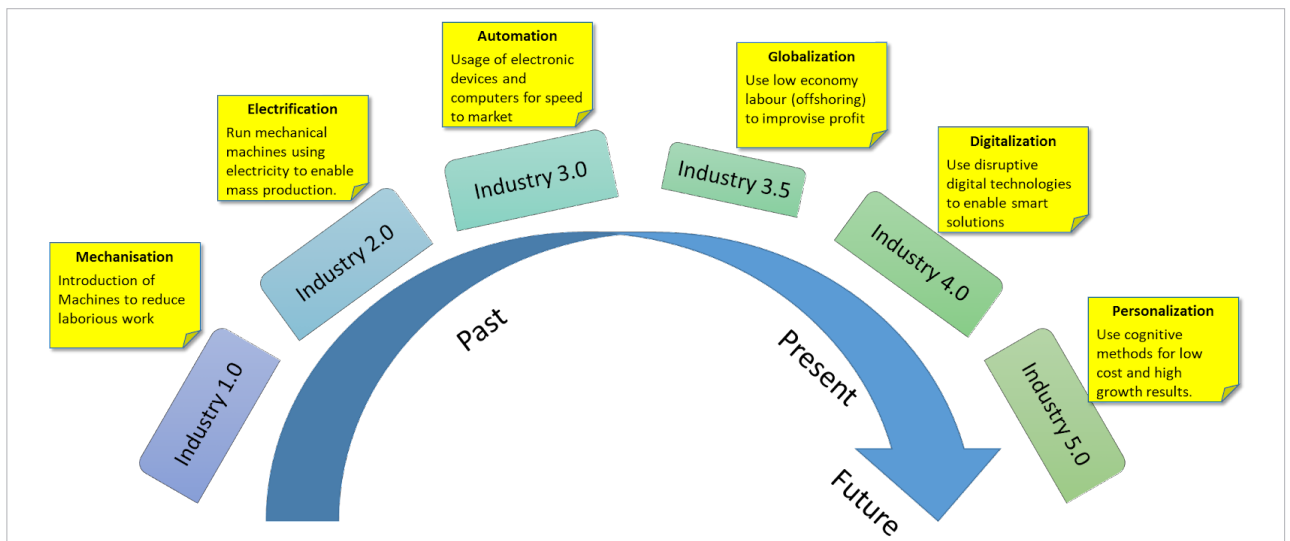


Figure 1: Evolution cycle of Industry 5.0

these simplifies some complex business and technical requirements. Let's explore these terms.

- **Internet of Medical Things (IoMT)** – Commonly used for technology solutions combined with sensors for health monitoring and remote patient monitoring in the healthcare industry
- **Internet of Drone Things (IDT)** – Utilises drones and unmanned aerial vehicles (UAV) for applications like agriculture, wildlife monitoring, evacuation drills, and industrial site monitoring
- **Internet of Digital Twins (IoDT)** – Replicates human-like virtual assistants, robotic assistants and chatbots to aid end users, e.g., virtual banking assistants for financial advisors
- **Internet of Vehicle Things (IoVT)** – Applied in the automotive industry for digital metaspaces to sell cars, simulators for vehicle driving training, and vehicle design using digital canvases
- **Internet of Robotic Things (IoRT)** – Used in production design to create a low-cost simulated environment that uses robots for inspection, quality tests and training
- **Internet of Intelligent Things**

(IoIT) – Applied in industries like virtual/online retail stores to analyse user patterns and behaviour to provide intelligent services based on machine learning

Internet of Everything (IoE) combines all or some of these services to integrate various processes, using the internet for data handling, as well as specialised devices like head-mounted devices, IoT sensors and monitors, and people with defined roles/responsibilities to handle various business functions.

In simple terms, these services (IoE) are not confined to specific technologies but use various combinations to enhance customer experiences, provide better solutions, and reduce costs for 'go to market' solutions across different sectors.

Trends in IoT-enabled application architecture

The latest trends in IoT-enabled application architecture include:

- **Blockchain for IoT security:** Blockchain technology can secure IoT devices and data by providing a decentralised and tamper-proof system.
- **Smart home devices:** IoT-enabled smart home devices are gaining popularity, offering remote control

and communication with other household devices.

- **Cybersecurity priority:** With the growing prevalence of IoT devices, cybersecurity becomes increasingly important. IoT-driven software architectures must consider the implications of software complexity, security, and privacy.
- **Edge computing rising over cloud computing:** Edge computing is gaining prominence due to its ability to process data faster, reducing the need to transmit data to the cloud for processing.
- **AI and Big Data significance:** IoT is a key source of real-time data for AI applications, enabling AI-driven decisions.
- **Complex wearable devices:** IoT-enabled wearable devices used for health monitoring and fitness tracking are becoming more complex.
- **Microservices architecture adoption:** IoT-centric software architecture poses unique challenges, and adopting microservices architecture can help.

Overall, IoT-enabled application architecture must evolve to meet the unique demands and challenges posed by IoT, as emerging IoT devices become the new norm.